An XAI-Driven Investigation of Zero-Shot Domain Generalization in Neural Waveform Classification

XAI Project Report

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Abstract:

Deep learning systems require massive quantities of labeled data, yet in many scientific and biomedical settings—such as extracellular neural waveform classification, annotations are scarce, expensive, and highly imbalanced. Hybrid learning, which combines easily generated synthetic data with limited real-world recordings, offers a promising solution. However, our initial attempts at zero-shot transfer failed: models trained exclusively on biophysical simulations achieved excellent in-domain performance but near-chance generalization on real recordings. This report presents an explainability-driven investigation of that reality gap. Using a proof-of-concept pipeline for multichannel extracellular spike footprints classification, we combine interpretable linear models, attribution methods, representational similarity analysis (CKA), and domain-adversarial training to diagnose why synthetic-to-real transfer breaks down. First, we reveal that baseline models exploit a temporal misalignment shortcut rather than learning meaningful waveform morphology. Second, we demonstrate that convolutional models trained on different domains rely on divergent salient regions and hidden representations, even when downstream zero-shot accuracies appear similar. Third, we show that unsupervised domain adaptation via a Domain-Adversarial Neural Network (DANN) is strictly architecture-dependent. While low-capacity models remained unstable, high-capacity residual networks successfully bridged the domain gap, improving zero-shot classification accuracy on real data from near-chance (35%) to 79%, closely approaching the fully supervised ceiling (85%). Ultimately, this study argues that Explainable AI (XAI) is not merely a post-hoc visualization tool, but a mandatory diagnostic framework. By enabling qualitative, quantitative, and visual sanity checks, XAI provides the critical feedback necessary to guide hybrid learning pipelines toward robust, domain-invariant representations.

A Colab notebook of this study is available.

1 Introduction

Supervised deep learning has achieved impressive performance across many application domains, but its success is tied to the availability of large, high-quality labeled datasets. In practice, this requirement becomes a major limitation in fields where annotation is expensive, slow, invasive, or dependent on expert judgment. Neuroscience is a clear example: extracellular neural recordings can be collected at scale, yet reliable cell-type labels remain scarce, uncertain and highly imbalanced. This mismatch motivates the use of hybrid learning, where models are trained from a combination of synthetic and real data.

Synthetic data are attractive because they can often be generated cheaply, at scale, and with automatic labels. In domains such as robotics, computer vision, and neuroimaging, simulation has become a major source of supervision, especially when real annotations are rare or costly [1]-[2]-[3]-[4]. Synthetic

datasets also offer practical and ethical advantages: they can reduce privacy constraints, avoid risky or inaccessible experiments, and make rare conditions easier to study [1].

Hybrid learning is therefore appealing not only as a data augmentation strategy, but as a route toward *zero-shot* or *few-shot* transfer. Ideally, a model trained on simulated waveforms would learn physiologically relevant features and generalize directly to real recordings. In practice, however, this ideal is hard to achieve because synthetic and real domains differ in subtle but consequential ways. This discrepancy is often described as a *reality gap* or, more broadly, a *domain shift* [2]-[5]

Many approaches have been developped to bridge domain differences, from simple domain randomization [6] to more complex and modern (un)supervised domain adaptation methods. Yet these approaches do not guarantee success when the gap involves hidden acquisition artifacts, temporal alignment errors, or structural differences in the signal itself. These challenges are particularly important in our setting: classification of multichannel extracellular neural waveforms using synthetic data from detailed biophysical simulations and real data from the C4 dataset [7]. Previous work in this pipeline showed that models trained only on synthetic data failed to generalize to real recordings, despite strong in-domain performance, both for simple methods and modern domain adaptation such as unsupervised domain adversarial neural networks (DANN) [8] or harmonization of datasets with VAE-GAN [9]. Yet, human eyes were able to find similitudes between real and synthetic classes, meaning the synthetic data should not be useless.

For that reason, this work adopts an explainability-first diagnostic strategy. Rather than treating explainable AI (XAI) as an optional interpretability layer added after training, we use it as an investigation framework for model failure. The goal is not only to ask whether a model performs poorly, but why. Does it rely on meaningful waveform morphology, or on acquisition artifacts? Do real-trained and synthetic-trained models attend to similar regions? Do they learn compatible hidden representations, or entirely different internal taxonomies? Can adversarial adaptation actually align domains without destroying semantic structure? To answer these questions, we designed a notebook organized around three phases: interpretable analysis of domain discrepancy, explainable analysis of zero-shot prediction, and explainable analysis of domain-adversarial training.

2 Problem Setting and Data

The proof-of-concept study focuses on classification of extracellular action-potential footprints. Each sample is represented as a short spatiotemporal tensor of shape $(C, H, W) = (1, 9, 180)$, corresponding to 6 ms of signal sampled at 30 kHz across a local neighborhood of 8 electrodes around the electrodes with the highest signal (called the main electrode). Two feature types are considered. MWF retains the multichannel spatial layout of the waveform across neighboring electrodes. 2DWF preserves the same input dimensions but repeats the main electrode across channels, thereby suppressing informative spatial variation, it theoretically contains less information, but better signal-to-noise ratio.

The target domain consists of real extracellular recordings from the C4 dataset from [7], where labels are sparse, partially available, and strongly imbalanced. The source domain consists of synthetic waveforms generated with LFPy simulations, internally made, available only on request. For the present investigation, target labels are deliberately excluded during unsupervised domain adaptation training to mimic a zero-shot synthetic-to-real transfer scenario. They are used only for evaluation.

This setup is representative of a broader scientific problem: rich simulation-based supervision on one side, scarce and noisy real labels on the other.

For more details on the data, a Colab notebook is available.

3 Methods

3.1 Phase 1: Interpretable analysis of domain discrepancy

The first phase addresses a basic but crucial question: what makes the synthetic and real domains separable? In previous experiments, Domain-Adversarial Neural Network (DANN) training often failed because the domain discriminator consistently outperformed the feature encoder, suggesting that the two domains remained too easy to distinguish. This indicated that the synthetic-to-real gap was likely too large, preventing stable adversarial alignment. Therefore, before training deep models, we first used simple interpretable methods, namely Logistic Regression (LR) and Principal Component Analysis (PCA), to determine whether domain separation relied on obvious shortcuts.

To characterize this baseline domain discrepancy, we trained a simple LR classifier to predict the domain label, i.e. source (synthetic) versus target (real). Because LR is directly interpretable, its learned weights can be visualized to reveal which temporal or spatial regions of the waveform contribute most strongly to domain classification. In parallel, PCA was used to assess whether a small number of dominant components captured the main axes of variation between the two domains.

3.2 Phase 2: Explainable analysis of zero-shot prediction

The second phase examines why models trained on one domain fail on the other. Two architectures were compared: a lightweight SimpleCNN and a deeper CustomResNet. Each model was trained on a single domain only, either real or synthetic, then evaluated on both domains to assess zero-shot transfer.

This phase combined three levels of analysis. First, standard supervised metrics measured whether a model trained on one domain could classify the other. As expected, both architectures showed poor zero-shot performance relative to their in-domain accuracy. Second, local explanation methods were used to inspect which input regions supported individual predictions. Five attribution families were selected to provide complementary viewpoints: Saliency, Integrated Gradients, Integrated Gradients with Noise Tunnel, DeepLift, and Grad-CAM. The goal was not to benchmark XAI algorithms exhaustively, but to use all five allows to achieve consensus-based qualitative explaination. If all attributions map toward noisy regions of the images such as corners high-frequencies, then the models are not focusing on relevant features. Third, we quantified global similarity of learned representations using Centered Kernel Alignment (CKA) [10]. Unlike visual inspection of attribution maps, CKA measures whether two models organize samples similarly in hidden space. This is especially appropriate here because the number of samples is much smaller than the number of parameters, a setting where older similarity measures such as CCA can become unreliable. CKA was computed across layers for models trained on different domains, across architectures within the same domain, within the same model across layers, and per class at the final representation layer.

3.3 Phase 3: Explainable analysis of domain-adversarial learning

The third phase investigates whether unsupervised domain adaptation can repair the divergence observed in Phase 2. We implemented a DANN with a shared encoder, a source-label classifier, and a domain discriminator connected through a gradient reversal layer. Source data were labeled synthetic waveforms; target data were unlabeled real waveforms.

This phase was again analyzed with more than accuracy alone. In addition to classification metrics, we tracked discriminator performance, gradient norms, embedding evolution through dimensionality reduction, and distribution alignment through Maximum Mean Discrepancy (MMD). CKA was then reused to compare DANN representations with those learned by real-only and synthetic-only baselines. This was important because successful adaptation should not only increase target accuracy, but also produce representations that are interpretable and semantically coherent.

4 Results

4.1 Phase 1

Contrary to our initial hypothesis that the model would separate domains based on baseline amplitude discrepancies, weight analysis revealed the exploitation of a massive temporal misalignment shortcut. The LR model achieved 100% domain separation accuracy simply by evaluating a specific temporal phase shift; synthetic spikes occurred earlier than real ones.

PCA confirmed this: aligning the waveforms caused the variance explained by PC1 to drop from 69.2% to 44.9%, indicating that temporal mismatch previously dominated the dataset's variance. Even after enforcing strict structural alignment, the LR model retained 97% domain classification accuracy due to geometric discrepancies, such as the decaying phase of the action potentials.

Implications for Pipeline Design: This phase established that standard translation-invariant models (like CNNs) would fail because they easily latch onto high-frequency domain artifacts. Consequently, the preprocessing pipeline required an aggressive update, integrating random temporal translations to destroy the phase-shift shortcut, alongside heavy Gaussian blurring and low-pass filtering to obscure domain-specific high-frequency noise.

4.2 Phase 2

Across both CNN and ResNet-like models, training on a single domain yielded strong performance on the seen domain but poor performance on the unseen one. The ResNet performed slightly better overall, suggesting that some additional nonlinear structure may be useful, but the gain was limited. Models trained on real data generalized somewhat better to synthetic than the reverse, which suggests that the synthetic domain may not capture enough of the variability present in real recordings. This asymmetry is consistent with a familiar simulation problem: synthetic data may be clean, structured, and labeled, but insufficiently diverse.

Initial attribution mapping revealed that models were still overfitting to spurious, non-informative regions (e.g., pre-trough noise). After revising the pipeline, especially the augmentation (adding random frequency filter and noise injection), zero-shot performance improved and attribution maps shifted toward more waveform-local regions, especially around the trough and repolarization phase. Example attribution maps are shown in Appendix, Fig. 5. However, the qualitative differences between domains remained informative. Models trained on synthetic MWF data tended to distribute attention more broadly across channels, suggesting reliance on smooth spatial decay patterns present in simulation. By contrast, models trained on real data produced narrower attention concentrated around the main channel, likely because the real spatial information had lower signal-to-noise ratio. This observation helps explain why 2DWF sometimes generalized better than MWF: removing spatial variability prevented models from overcommitting to synthetic-specific cross-channel structure.

While attribution maps offer local insights, they cannot confirm whether real-trained and synthetic-trained models learn the same internal taxonomy, but that exactly what CKA enables. CKA revealed a more nuanced picture than performance curves. For the CNN, models trained on real and synthetic data retained a relatively stable, block-like alignment across layers, suggesting that they learned comparable feature hierarchies despite poor transfer. For the ResNet, alignment was high in early layers but decayed progressively in deeper layers. This implies that both models initially extract similar low-level signal statistics, but the deeper ResNet transforms them into increasingly domain-specific representations. Despite achieving similar downstream accuracy, the ResNet transformed the synthetic signal into a deep representation completely alien to the real-trained model, indicating an exploitation of simulation-specific features. So, two models can display similarly weak zero-shot accuracy, yet fail for different reasons. The CNN appears more constrained and therefore more representationally stable. The ResNet, with higher capacity, appears more able to exploit latent synthetic-specific shortcuts.

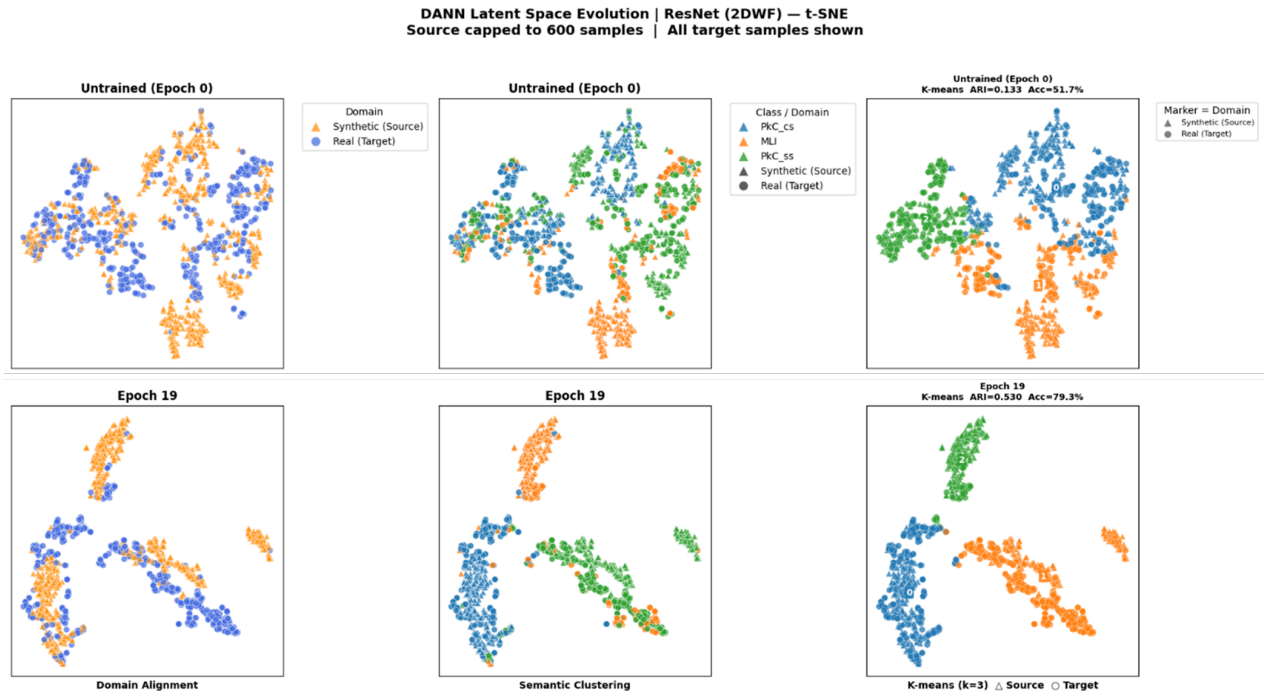

Figure 1: Successful unsupervised domain adaptation with the ResNet-based DANN. Across training snapshots, the source and target embeddings progressively overlap while preserving semantic structure between neuron classes. The right-most panels show that unsupervised clustering in the adapted latent space recovers target classes well above chance, supporting the claim that the learned representation became both more domain-invariant and more semantically organized.

4.3 Phase 3

Domain-adversarial training did not behave uniformly across models. For the low-capacity CNN, the discriminator often overpowered the encoder, leading to unstable learning and little improvement over naive zero-shot baselines. After significantly reducing the latent space dimension, the discriminator remained stable, and the DANN-CNN performed better, but its embeddings remained very noisy (Appendix, Fig. 3). For the ResNet, DANN produced clearer gains on unseen real data and generated embeddings in which source and target samples overlapped more visibly while preserving class structure (Fig. 1).

A particularly encouraging result was that simple K-means clustering in the adapted latent space achieved much better-than-chance performance on target data, indicating that the learned representation had become more semantically organized even without target labels during training. However, nothing guarantee that the latent space is gaussian-like, hence we tried graph-based clustering that rely on local proximity, because the optimizer impose regularization to get smooth weights. Spectral clustering achieved up to 79.3% accuracy on zero-shot classification. This is fairly good results knowing that similar architecture was performing near change-level (33%) at the beginning, and that the same model, fully supervised, achieves $\sim 85\%$, as illustrated in Appendix, Fig. 2.

At the same time, two quantitative diagnostics behaved paradoxically. First, MMD sometimes increased across training despite visually improved manifold overlap. This could be due to the fact that the initial distribution of embeddings is a messy blob, mixing everything, where each data is close to each other, while DANN push class apart from others as we see in Fig. 1.

Second, DANN-trained ResNet representations showed low CKA similarity to real-only trained ResNet representations. Rather than interpreting this as failure, we argue that it may reflect a desirable property: the real-only model was trained on very limited labeled data and may therefore overfit to recording-specific artifacts, whereas DANN forces the representation away from those artifacts and toward more transferable geometry. These results support a cautious conclusion: DANN can help,

but only under favorable architectural conditions, and classical alignment metrics do not always track adaptation quality faithfully.

5 Discussion and Limitations

First, while hybrid learning is promising, it is inherently fragile. Synthetic data are highly valuable because they provide abundant, clean supervision in settings where real-world labels are scarce. This makes the approach relevant not only in neuroscience but also in robotics, medical imaging, and autonomous systems where annotation is costly or dangerous. However, this study demonstrates that successful transfer depends on more than simply augmenting datasets with synthetic examples; it requires ensuring that models cannot exploit synthetic-specific artifacts at the expense of learning domain-invariant physiological features.

Second, architectural capacity dictates the success of domain adaptation. While both the low-capacity CNN and the high-capacity ResNet exhibited poor zero-shot generalization initially, they failed for different reasons, as revealed by CKA. Furthermore, during adversarial training, only the higher-capacity ResNet possessed the representational flexibility required to navigate the adversarial objective and successfully map the source and target domains into a shared latent space. This difference is also visible in the appendix figures: the CNN failure case remains noisy after adaptation (Fig. 3), whereas the self-CKA matrices show that the CNN learns a more blocky, weakly evolving internal representation while the ResNet transforms features more progressively across depth (Fig. 4).

Third, XAI is essential as a diagnostic debugging instrument. In this project, explainability was not used to justify a final model to an end-user, but rather as a scientific inspection framework. A simple, interpretable domain classifier exposed the temporal misalignment shortcut that was likely causing previous iterations of the DANN to collapse. Attribution maps highlighted model reliance on noisy, non-informative regions, prompting critical revisions to the data augmentation pipeline. CKA revealed that similar downstream accuracies concealed fundamentally different internal logics, and embedding visualizations confirmed when domain alignment genuinely improved. Furthermore, these tools uncovered critical implementation errors, such as missing standardization in the evaluation datasets. This serves as a strong argument for integrating XAI early in experimental workflows, particularly in transfer learning studies where failure modes are subtle and not captured by accuracy metrics alone.

Several limitations must be acknowledged. Primarily, this study serves as a proof-of-concept rather than a fully controlled, exhaustive benchmark, which remains a necessary next step. Due to the high computational cost and extensive hyperparameter space of the models, some analyses rely on qualitative inspection and selected examples rather than strict statistical confidence intervals across repeated runs. Second, because the emphasis was diagnostic, models were not strictly evaluated using a classical train/validation/test split within each domain; therefore, performance on the seen domains should not be overinterpreted (especially in phase 1 and 2). Third, XAI methods provide complementary but imperfect views of model behavior. Attribution maps remain highly method-dependent, and apparent cluster alignment in 2D dimensionality reductions (e.g., UMAP, t-SNE) does not guarantee semantic consistency in the full latent space. Finally, the synthetic dataset may still underrepresent the true diversity and noise characteristics of the real-world domain, as we intentionally restricted the analysis to the three most realistic classes out of the five available.

6 Conclusion

This report investigated why zero-shot synthetic-to-real transfer fails in neural waveform classification, despite the theoretical promise of hybrid learning. The primary conclusion is that domain generalization failure cannot be fully understood or resolved through performance metrics alone. Before targeted adaptation, the source and target domains were separable via trivial temporal shortcuts. Following initial training, models operating on different domains relied on disparate salient regions and developed

fundamentally divergent hidden representations.

By utilizing XAI to curate the pipeline and prevent the exploitation of these mathematical shortcuts, we enabled Domain-Adversarial training to successfully recover performance, albeit in an architecture-dependent manner. Ultimately, zero-shot learning with models trained exclusively on synthetic data was improved from near chance-level to 60% accuracy, and the application of DANN pushed this to 79%, closely approaching the theoretical 85% ceiling of fully supervised models.

The broader lesson is that successful hybrid learning requires more than just synthetic supervision and standard domain adaptation recipes. It demands careful control of nuisance cues, explicit reasoning regarding feature invariance, and systematic, explainability-based sanity checks. In this context, XAI is not an optional appendix to domain adaptation research, but a fundamental prerequisite for robust methodology.

References

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A Appendix

A.1 Additional figures

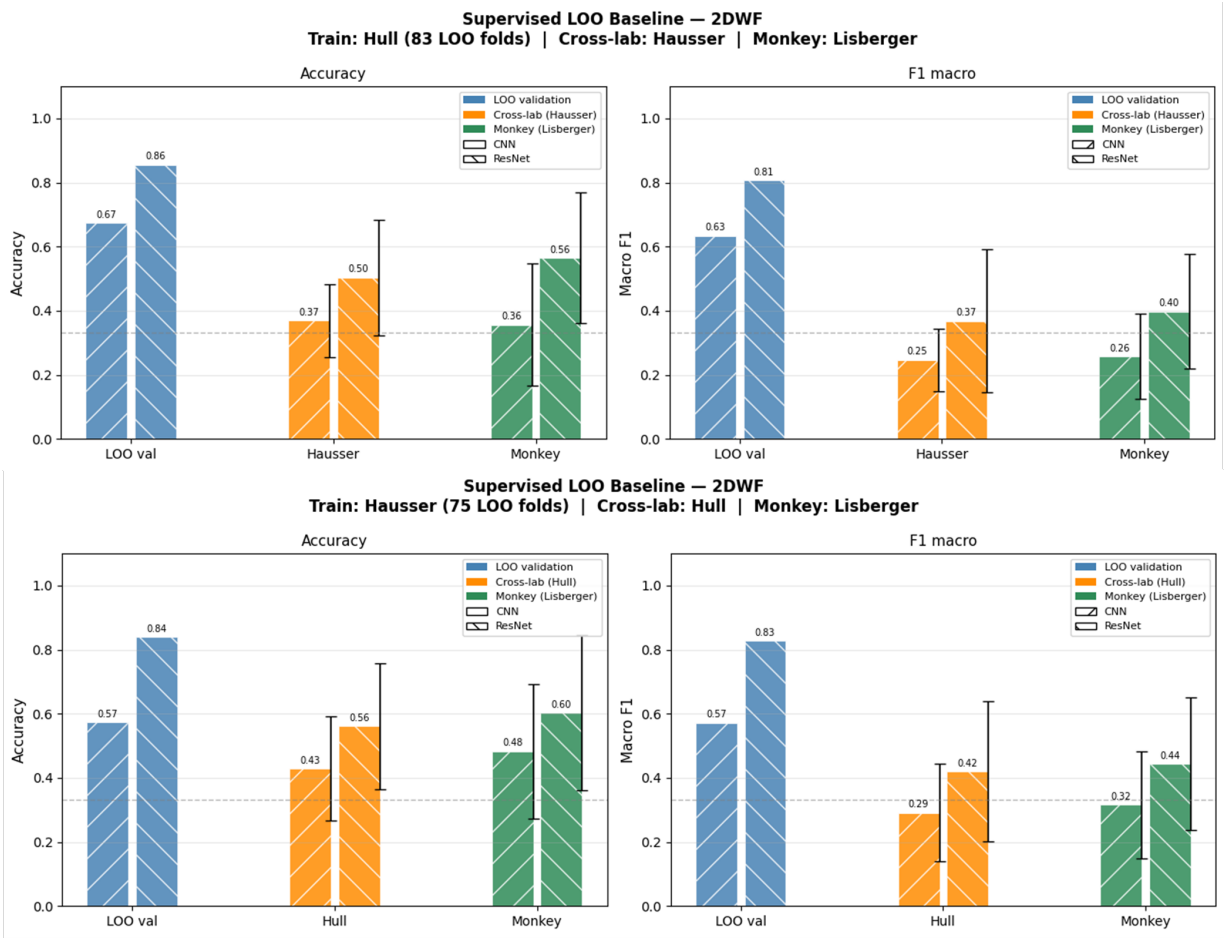

Figure 2: Baseline supervised training and zero-shot evaluation curves before domain-adversarial adaptation. Both architectures achieve strong in-domain performance but weak cross-domain transfer, motivating the need for a dedicated XAI-based diagnosis of failure modes.

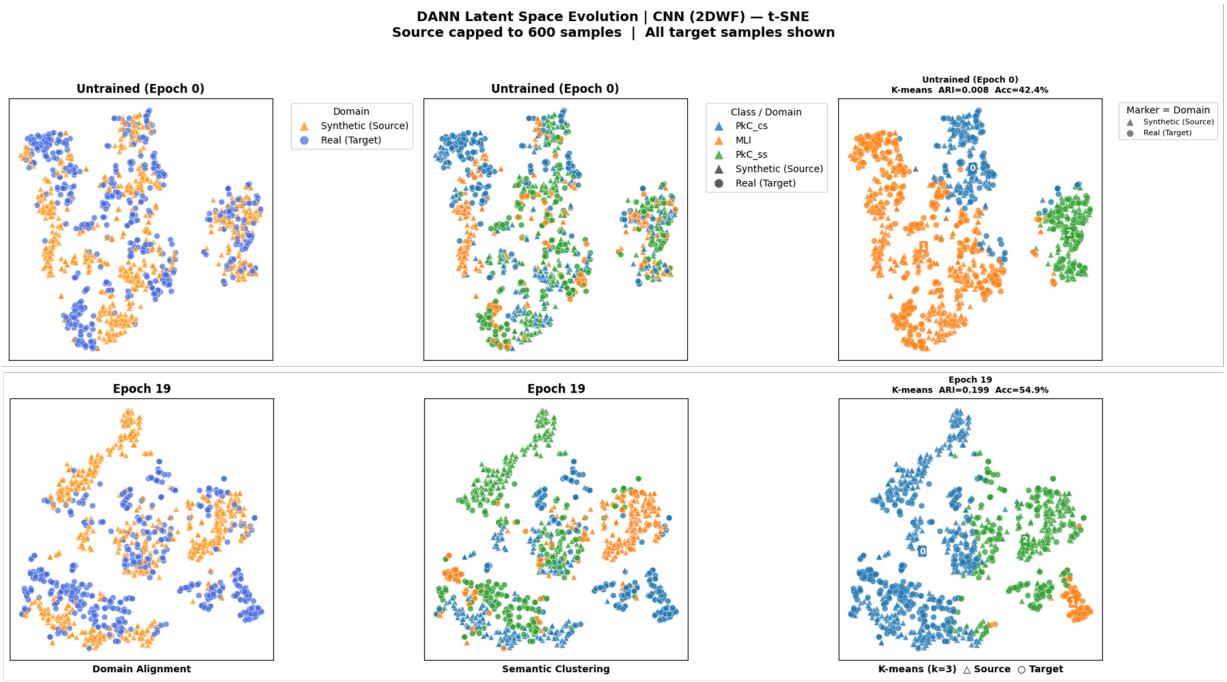

Figure 3: Failure mode of the CNN-based DANN. Even after stabilizing adversarial training, the learned latent space remains noisy and weakly structured, suggesting that the low-capacity encoder lacks sufficient flexibility to jointly enforce class discrimination and domain invariance.

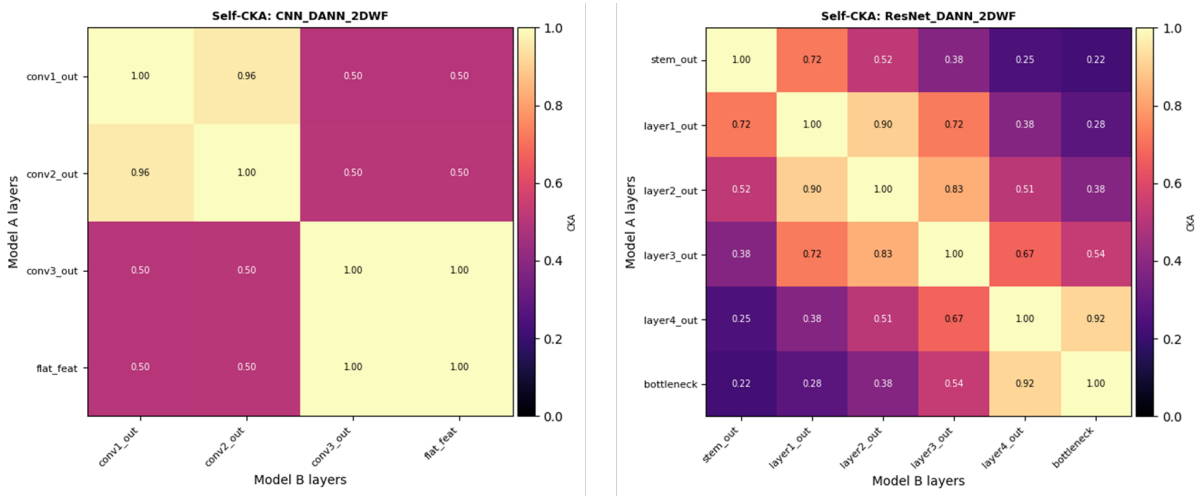

Figure 4: Within-model self-CKA comparison. The CNN exhibits a blocky structure, indicating that its internal representations change relatively little across depth. By contrast, the ResNet shows a more progressive, almost linear layer-wise evolution, consistent with deeper hierarchical transformation of the signal representation.

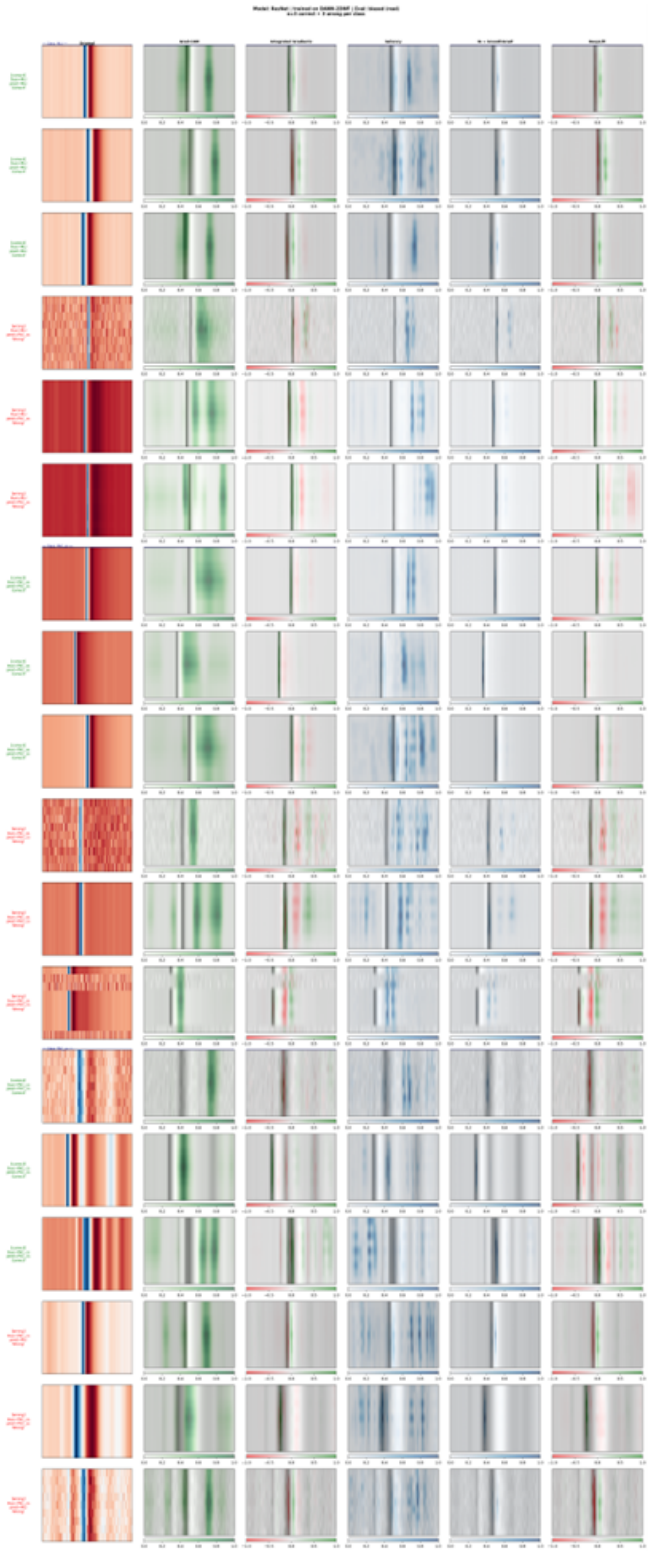

Figure 5: Representative attribution maps used for qualitative model inspection. These examples illustrate how the explanation toolkit helped identify spurious focus on non-informative regions and motivated revisions of the preprocessing and augmentation pipeline so that models attend more strongly to waveform-local regions around the trough and repolarization phase.